

Incentivizing Recycling Behavior: Repro - A Reward based CNN-Enabled Framework for Smart Waste Collection

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Abstract—Efficient waste management together with certain measures to promote the recycling process are essential for overcoming global environmental issues. In this study, we describe Repro an app, which is targeted at transforming how we collect waste and recycle by integrating CNN technology and a reward-based system. Users, with the Repro app, take pictures of any trash they discard and rely on the underlying CNN model to identify the type of trash correctly. If a user's trash accumulation reaches beyond a given limit the system sends a notification to the trash collectors via the riders app to direct for a timely waste collection. User tokens are provided to users after trash collection, which can be redeemed within the Repro app for various product items and coupons. Moreover, this motivating approach encourages responsible waste management which goes along with the development of environmental custodianship among users. We describe the technical implementation of the CNN-based framework, assess the performance of the level of waste classification accuracy, and predict the behavioral implications of the reward system on recycling behavior. The study illustrates that Repro is a scalable solution for smarter waste collection and recycling endeavors, consequently creating a more eco-minded and sustainable environment.

Index Terms—Smart Waste Collection, Convolutional Neural Networks (CNN), TensorFlow Lite, Mobile App, Incentive Mechanisms, Sustainable Consumption

I. INTRODUCTION

Effective waste management and promoting sustainable recycling practices have become imperative in mitigating environmental degradation. Our research focuses on developing innovative solutions by using technology to solve environmental problems. Specifically, we introduce Repro, a mobile application designed to streamline waste collection processes through advanced image recognition technology and incentivized recycling.

The global waste crisis poses significant challenges to ecosystems, public health, and resource sustainability [1]. By encouraging recycling behavior and optimizing waste collection methods, we aim to alleviate the burden on landfills, reduce pollution, and conserve natural resources. Our work addresses a pressing societal need for more efficient and environmentally conscious waste management practices.

In the face of escalating environmental concerns such as climate change and resource depletion [2], there is an urgent call for innovative solutions to foster sustainable living.

In recent times, technology, especially artificial intelligence [3]–[5], has gained a lot of prominence, thanks to the multitude of successful applications [6] in various fields. In addition to multiple optimization problems [7]–[9], the subfield of machine learning [10], [11], especially the part based on neural networks [12], [13] and deep learning [14]–[16] have proven to be successful in computer vision [17] and natural language processing [18], [19] fields. By harnessing the power of deep learning models, specifically MobileNet and behavioral economics, we can enable individuals to make eco-friendly choices in their daily lives. Our research not only addresses immediate environmental challenges but also contributes to building a more resilient and environmentally aware society.

A. Literature Review

Nowakowski et al. [20] proposed a smartphone app leveraging user-submitted photographs to categorize waste, particularly e-waste, for collection by organizations. This approach gives positive response for streamlining e-waste collection and recycling. Saptaputra et al. [21] built upon this concept, demonstrating how smartphone-based waste sorting can be adapted for household waste management. Furthermore, user interface design plays a crucial role in promoting user engagement with waste sorting activities, as evidenced by studies exploring the impact of user-friendly interfaces on increasing recycling activity.

To address the limitations of traditional incentive-based recycling systems, Zhou et al. [22] propose the use of "smart" incentive-based systems. They argue that the integration of Internet of Things (IoT) and data analysis technologies can significantly improve household recycling. Their proposed system utilizes four intelligent components – amount pattern discovery, price adjustment suggestion, collection amount forecasting, and information sharing – to enhance the effectiveness of incentive-based recycling programs. The importance of incentive based recycling can be seen from Betul et al

[23] about how almost half people were inclined to recycling purely based on the incentives that they were provided. It also highlights how awareness about recycling plays a crucial role in promoting a pro-recycling community.

B. Gap Analysis

One notable gap in the field of waste management and recycling is the limited utilization of both the deep learning models and the incentivizing of recycling practices to actively encourage the general public to recycle more. Although deep learning has been utilized for classifying waste materials and modifying waste management processes, there are limited businesses or startup companies that have applied this technology to influence an individual's behavior and to promote recycling by providing incentives to people who engage in recycling. There is a void here that can be filled by the development of a business or startup using the power of deep learning and incentives to motivate and encourage people to recycle and thus contribute to tackling the issue of waste management, as well as to building a better future.

C. Main Contributions

Waste Classification Model: We developed a highly precise and accurate waste classification model using a convolutional neural network (CNN). This model simplifies the waste sorting process, accurately identifying recyclable materials, and hence, plays a crucial role in bridging the gap between individuals and recycling services.

- We designed and implemented a unique token-based reward system within Repro. This incentive scheme motivates users to actively participate in recycling activities by rewarding them with tokens that can be redeemed in our online store.
- We created the Repro mobile application, integrating the waste classification model and the incentive scheme to provide a comprehensive solution for waste management. The application not only simplifies recycling but also makes it a rewarding experience, promoting long-term sustainable habits among users.
- Our study takes a intuitive approach to waste management, addressing key aspects such as user engagement, waste classification accuracy, and logistical efficiency. This comprehensive strategy sets our work apart and contributes to creating a more sustainable environment.
- We actively connected individuals and businesses with appropriate recycling services, facilitating efficient recycling practices. This initiative aids in reducing environmental harm and encourages active participation in recycling efforts.

In this research, we emphasized the importance of waste management and propose leveraging deep learning to enhance recycling practices. Our research highlights the need for effective solutions and introduces a recycling application utilizing image recognition technology to connect users with appropriate recycling companies and create a huge economic impact [24]. Overall, our findings underscore the urgency

of sustainable waste management and the potential of deep learning in promoting environmental preservation through optimized recycling.

II. PROBLEM STATEMENT

The diversion of water from rivers, waste in waterways, and pollution from landfills result from improper waste disposal and ineffective recycling methods which hurt the quality of natural habitats. A barrier to this scenario is the very inadequate state of the waste management and recycling sector that is being hampered by the lack of information and there is restricted access to this area. Many recyclable materials end up in landfills, adding to environmental harm. Furthermore, there is a considerable gap in connecting individuals and businesses with suitable recycling services, which is compounded by a lack of effective incentivization to encourage recycling. The challenge is to devise a solution that not only classifies waste accurately but also simplifies the recycling process, making it more accessible, and effectively motivates participation. Such a solution needs to address user engagement, improve waste classification, and optimize logistics to bring about substantial improvements in waste management practices.

III. METHODOLOGY

In this study, our aim revolves around the broader goal of making the world a cleaner and better place by promoting sustainable waste management practices and reducing the environmental impact of waste. By examining these research questions, we seek to contribute to the development of effective strategies and innovative solutions that foster greater participation in recycling efforts, thereby leading to a more sustainable and environmentally conscious society.

A. Data Acquisition and Description

The foundational data for this study is derived from a rich and diverse set of 5000 images, collated from established waste datasets named "waste-datasets-review" and "trashnet", uploaded on GitHub. The collected images span various categories of waste materials including Aluminium, Cartons, Glass, Organic Waste, Other Plastics, Paper, Cardboard, Plastic, Textiles, and Wood which are shown in Figure 1.



Fig. 1. Different types of wastes that RePro can identify.

These distinct categories provide the ground truth or reference points crucial for both the training and evaluation phases of the machine learning model dedicated to waste classification. This varied composition of waste types in the dataset bolsters the development of a comprehensive and precise classification system. Some sample images from the training data are shown in Figure 2



Fig. 2. Sample images from the training dataset.

B. Model Development and Integration

The waste classification model, a pivotal element of our research, was meticulously constructed using Python. The model used the training dataset for training and validation. Upon successful training and optimization, the model was transformed into a TensorFlow Lite format, facilitating its seamless integration into a mobile application developed with Kotlin. The mobile application, apart from hosting the machine learning model, provides a user-friendly interface for users to upload images of waste for classification. In a bid to promote recycling, a token system was incorporated within the application. Users accumulate tokens for every recyclable item submitted, which can subsequently be redeemed in an in-app store. Additionally, a rider app is made also in Kotlin for efficient waste collection from users. The overall flow followed in this study is shown in Figure 3.

Images of the interface for RePro user app are shown in Figure 4.

C. Neural Network Architecture and Training

The proposed model architecture capitalizes on transfer learning, tapping into the knowledge of a MobileNetV3 [25] Large base model pre-trained on ImageNet [26], which we then tailor to address our specific task by customizing the classification head. Operating with input images sized at 256x256 pixels and three color channels, we preserve the previously learned information by freezing all but the last six layers of the base model. Our classification head incorporates essential components such as a dropout layer (set at a dropout

TABLE I
NETWORK CONFIGURATION OF FCN USED

Network Configuration	
Epochs	50
Learning rate	0.00125
Dropout rate	0.45
Optimizer	Adam
L_1 Regularization	0.045
Samples in training set	4571
Samples in validation set	507

rate of 0.45), global average pooling, and batch normalization, followed by a dense layer (256 units) employing ELU activation and L1 regularization (strength = 0.045) to curb overfitting. A subsequent dropout layer is applied before a dense layer outfitted with softmax activation for class prediction. With an Adam optimizer (learning rate = 0.00125) and sparse categorical cross-entropy loss, our model undergoes training for 50 epochs, with learning rate scheduling callbacks dynamically adjusting the learning rate every epoch or every 6 epochs by a factor of 10 or 0.1, respectively. Evaluation of model performance hinges on accuracy, with the training dataset consisting of 4571 samples and the validation dataset comprising 507 samples, both utilized for rigorous assessment. The architecture of the CNN is shown in Figure 5 and a detailed depiction of the hyper-parameter settings can be found in Table I.

IV. RESULTS

The waste classification model trained on the dataset of 5,000 images achieved an overall accuracy of 98.09% percent in accurately categorizing different types of waste. The model exhibited high precision and recall scores across multiple waste categories, such as aluminum, cartons, glass, organic waste, other plastics, paper and cardboard, plastic, textiles, and wood. This indicates the effectiveness of the machine learning model in bridging the gap between the general public and waste recycling. The integration of the TensorFlow Lite model into the mobile application allows users to conveniently upload images of waste items, and the model rapidly classifies the items, providing instant feedback and enabling users to recycle correctly.

The matrix 6 showcases the accuracy of the model in correctly classifying different types of waste materials. It offers insightful information on the model's true positive, true negative, false positive, and false negative predictions. The picture clearly states that our proposed technique categorizes garbage correctly, which helps make better waste management plans. The confusion matrix in Figure 6 provides effective criteria for the evaluation of the quality of the CNN model and for providing directions for further developments in garbage classification systems.

The implementation of the token system within the mobile application will prove to be a successful strategy for incentivizing recycling. Users will receive tokens based on the recycled items they submit through the app. With these tokens, the users will be able to access rewards and discount coupons in the in-app

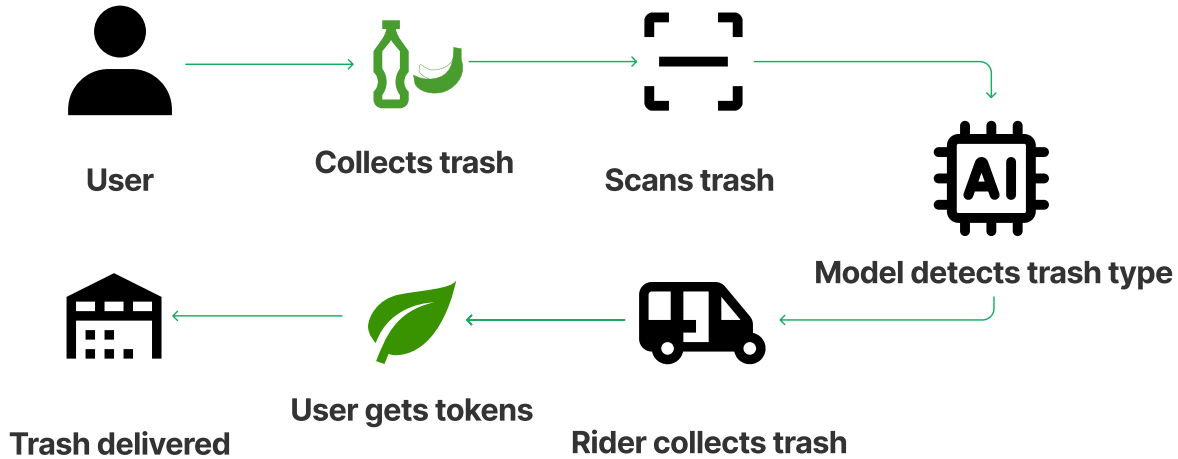


Fig. 3. Flow diagram of the entire process.

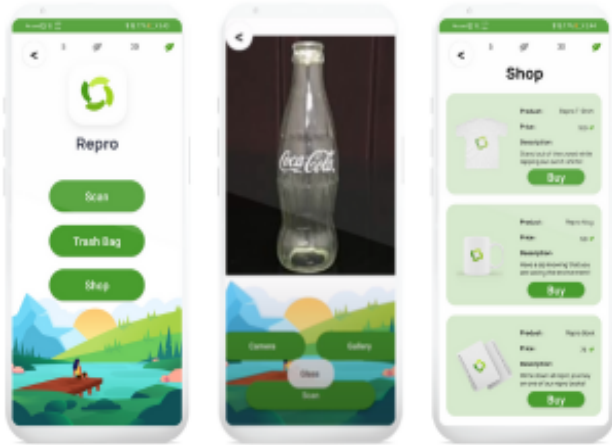


Fig. 4. Pictures of the App Interface.

store. This user incentive system will significantly contribute to an increase in recycling participation and motivation among app users. The overall outcome of this research is a tangible step towards achieving a more sustainable and eco-friendly environment by promoting responsible waste management practices.

The Python library Matplotlib [27] is used to construct a graph showing the progression of loss and accuracy values over the course of the epochs after the training is complete. This enables us to examine the model's behavior throughout training. The graph on the left shows the improvement in accuracy during the first epochs for both the training and test values (each dataset is represented by a separate line). However, a point is reached where the test accuracy falls below the training accuracy (depicted by the blue line), indicating overfitting. The larger the disparity between the training and test values,

the more pronounced the overfitting becomes, resulting in decreased generalizability of the model. Nonetheless, in this particular case, the discrepancy is minimal, with a difference of less than 1%. The training accuracy stands at 98.75%, while the testing accuracy is 98.09%. This slight variance does not significantly impact the model's outcomes.

V. DISCUSSION

In this way, we can realize that the mobile application with the integrated waste classification model will become the middleman for the public to recycle domestic waste. The accuracy rate of 98.09% is a clear manifestation of the system's precision and ability to differentiate between different types of waste. This indicates that the implementation of deep learning models along with user-friendly interfaces can be a driving force for people to contribute sustainably toward recycling. Instant feedback and guidance on correct waste classification are promoted by the application in order to encourage users to make informed decisions. Well-classified wastes are the goal and the consequence is contamination-free recycling streams.

The token system, together with the recycling app, will prove to be a useful means of inducing good recycling behavior. When tokens are provided as an incentive for recycling activities, it will engender community participation as the users get involved in the recycling efforts. This will in turn result in more people engaging with recycling and developing a positive attitude towards it. Gamification elements, including the token accumulating as well as redemption systems, will be grow user motivation and create feelings of satisfaction, which will, in turn, bring about a kind of sustainable recycling habits.

Overall, this study highlights the potential of deep learning and mobile applications in revolutionizing waste management and promoting sustainable recycling practices. By addressing

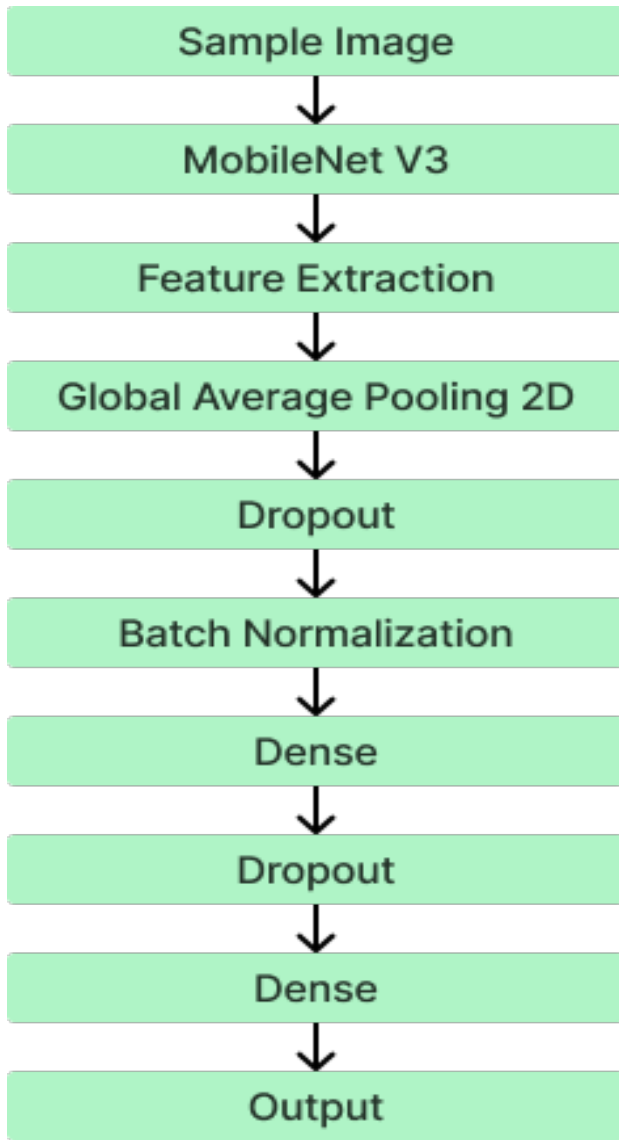


Fig. 5. Architecture of the CNN model for training and inference.

the gap between the general public and recycling companies, implementing user incentives, and optimizing logistical operations, we aim to pave the way for a more environmentally conscious society and contribute to global efforts towards a cleaner and better future.

VI. CONCLUSION

Through this research, we highlight the importance of our initiative, the Repro app, which will help to change the approach towards waste management and encourage people that recycling is an environmentally friendly way to get rid of household waste. We contribute our application with an innovative and advanced waste classification model, which is based on CNN techniques in order to exceed the current state of the art in waste classification and accuracy. This app, in turn, has shown to be vital in creating the right linkage between users and the most appropriate recycling companies with the

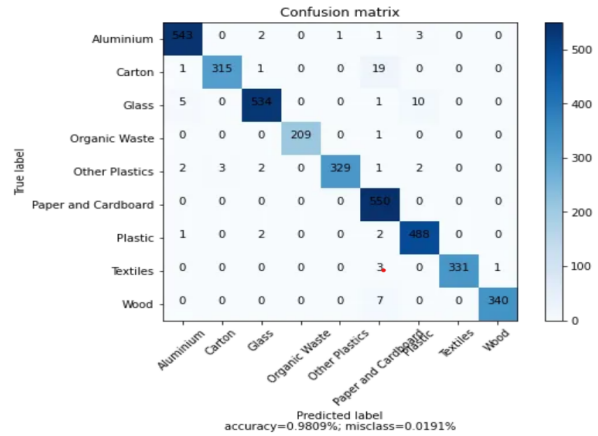


Fig. 6. Confusion matrix - Proposed waste classification model.

aim of acting as a channel that would connect the information and recycling accessibility gaps in the recycling sector. As another innovation, we go with a token-based reward system for our application. This reward system will essentially be the underlying driving force of users' recycling behavior. This will, on the other hand, not only initiate an active participation of their involvement in recycling but also promote commitment to sustainability. These tokens by their nature would reward the eco-conscious behaviors of individuals and enterprises, the task of recycling which in most situations is considered to be rather monotonous and boring, would get transformed into an exciting and rewarding activity.

The novelty of our work lies in its comprehensive and holistic approach, which simultaneously addresses three fundamental aspects: helping with user engagement, bettering waste classification, and providing logistic support. The latter will be achieved by means of our already-in-beta testing application for riders who will be a fundamental part of our system as far as the transformation of the pickup process is concerned. With this feature, recycling management can become convenient, and ultimately, the entire process would be smooth for people to get rid of their trash in a more nature-friendly manner. The study proves the use of e-technologies in the attainment of positive and measurable changes in solid waste management. The results underpin the capacity of such novel mechanisms to attain this through more responsible behaviors amongst each individual and business. It reveals that integration by the use of technology, drive, and logistics may help reach this objective. The projects we undertake in this regard are designed to change the perception of recycling from being an additional duty that we have to be a part and parcel of our daily routines, growing a more green and sustainable future.

VII. FUTURE WORK

Ahead Repro's, development strategy focuses on the following improvements that reflect our mission to promote sustainable living. We are interested to see if our incentive system will drive people to lighten their footprint via our

model by actually launching our app and making it accessible to the general public starting with Karachi which has the worst problem of waste management in Pakistan. Moreover, the app will feature real-time feedback, allowing users to grasp the environmental consequences of their recycling behavior, which will result in higher engagement with environmentally conscious behaviors. In addition, we would like to proceed with the expansion of our waste sorting model so that it is capable of comprehending an even wider range of waste disposables, consequently strengthening the system's performance within various recycling arrangements. Even further, we will seek partnerships with recycling organizations and local government agencies, to be able to tap into their expertise that could refine our recycling process, and serve as the basis of the development and sustainability of our recycling initiatives.

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